

Explainable Transfer Learning for Brain Tumor MRI Classification: A Four-Model Benchmark with Duplicate-Controlled Internal Testing and External Validation

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Abstract—Brain tumor classification from magnetic resonance imaging (MRI) is a clinically important computer vision problem, but many public-dataset studies rely only on internal test accuracy and do not sufficiently control image duplication or external generalization. This paper presents an explainable four-class brain tumor MRI classification benchmark for glioma, meningioma, no-tumor, and pituitary categories. The internal development cohort was constructed from the Masoud Nickparvar and Ishans24 Kaggle datasets, while the Mendeley 2026 Brain Tumor MRI Dataset was held out as an independent external validation source. A preprocessing and leakage-control pipeline detected 28,908 raw images, successfully preprocessed all images, identified 20,400 duplicate entries, removed duplicate overlap, and produced 15,721 final images with zero duplicate rows across train, validation, internal test, and external validation splits. Four models were evaluated under a unified PyTorch protocol: a scratch-trained Basic CNN and three ImageNet-initialized transfer-learning models, namely ResNet50, EfficientNetB0, and DenseNet121. All models were trained or fine-tuned as four-class classifiers, with controlled augmentation, class-weighted cross-entropy, internal test evaluation, independent external validation, confusion matrices, and Grad-CAM visual explanations. EfficientNetB0 achieved the highest internal test accuracy of 97.99%, while DenseNet121 achieved the strongest external validation performance with 87.79% accuracy and 87.95% weighted F1-score. ResNet50 was close on external validation with 87.42% accuracy and 87.58% weighted F1-score. The results show that external validation changes model ranking and that duplicate-control reporting is essential for credible medical imaging model evaluation.

Index Terms—Brain tumor classification, magnetic resonance imaging, transfer learning, external validation, DenseNet121, ResNet50, EfficientNetB0, Grad-CAM, medical image analysis.

I. INTRODUCTION

Brain tumors are among the most serious neurological abnormalities because delayed or incorrect assessment may affect treatment planning, surgical decision-making, and patient outcomes. MRI is widely used for non-invasive visualization of brain tissue, but manual interpretation requires expert radiologists and can be time-consuming. Deep learning can support computer-aided diagnosis by learning discriminative image features for tumor categories, but high reported accuracy on public MRI datasets does not automatically imply robust clinical generalization. A common weakness in public-dataset research is that models are evaluated only on internal splits generated from the same source distribution. If duplicate or near-duplicate images appear across splits, performance can be inflated. If a model is not evaluated on an independent external dataset, its real-world robustness remains uncertain.

This work addresses these issues through a duplicate-controlled and externally validated benchmark. We formulate a four-class classification task: glioma, meningioma, no-tumor, and pituitary tumor. The experimental design combines two public sources for internal development and reserves a third public source for external validation. Four architectures are compared: a Basic CNN baseline, ResNet50, EfficientNetB0, and DenseNet121. The selected transfer-learning models represent three widely used architectural design principles: residual learning, compound scaling, and dense feature reuse. To improve interpretability, Grad-CAM visualizations are reported for internal and external examples.

The main contributions are as follows:

- A duplicate-controlled preprocessing pipeline that harmonizes three public brain tumor MRI sources, removes

- duplicate overlap, and reports zero final split leakage.
- A four-model benchmark comparing a scratch CNN with ResNet50, EfficientNetB0, and DenseNet121 under the same four-class MRI classification protocol.
- Independent external validation on the Mendeley 2026 dataset to test generalization beyond the internal development cohort.
- A complete evaluation using accuracy, precision, recall, F1-score, confusion matrices, generalization gap analysis, and Grad-CAM explainability.

II. RELATED WORKS AND RESEARCH GAP

Deep learning methods have become central to medical image classification. ResNet introduced residual connections to enable very deep CNN training [4], DenseNet strengthened feature propagation through dense connectivity [5], and EfficientNet proposed compound scaling to balance depth, width, and resolution [6]. These architectures are frequently adapted for medical image classification because transfer learning can reduce the amount of task-specific training data required.

Recent brain tumor MRI studies continue to report strong internal performance using CNN, EfficientNet, transformer, and detection-based approaches. A 2025 Squeezed-Eff-Net study used the public Nickparvar dataset and reported high accuracy using an EfficientNet-B0-based hybrid model [8]. A 2025 YOLOv8/YOLOv11 study considered the four categories glioma, meningioma, pituitary tumor, and no-tumor using transfer-learning fine-tuning [9]. A lightweight CNN study in 2025 used a four-class public MRI dataset and reported 5-fold cross-validation for robust internal assessment [10]. In 2026, a Vision Transformer study on BRISC2025 again studied the four-class brain tumor problem and compared against CNN baselines [11]. These studies confirm the relevance of the four-class MRI task, but they also show a gap: many reports emphasize accuracy while giving less attention to duplicate leakage, dataset-source separation, and external validation.

The research gap is therefore not simply the absence of another high-performing CNN. The gap is the need for a reproducible comparison in which dataset sources are justified, duplicate overlap is removed, external validation is kept separate from model development, and model decisions are inspected with explainability maps.

TABLE I: Summary of Related Brain Tumor MRI Classification Studies and Remaining Gaps

Ref.	Year	Dataset/Task	Method	Reported Result	Limitation Relevant to This Work
[8]	2025	Nickparvar four-class MRI dataset	SqueezeNet + EfficientNet-B0 + handcrafted descriptors	98.93% test accuracy; 99.08% with TTA	Single public source; no independent external validation reported in the abstract.
[9]	2025	Four classes: no-tumor, glioma, meningioma, pituitary	YOLOv8/YOLOv11 and custom CNN	YOLO models above 99% accuracy in reported setting	Detection-style model; external dataset shift is not the central analysis.
[10]	2025	Public four-class MRI dataset	Lightweight CNN with tuning and 5-fold CV	98.78% classification accuracy	Strong internal validation, but not an independent external cohort study.
[11]	2026	BRISC2025 four-class dataset	ViT with colormap features	98.90% accuracy and 99.97% AUC	Different dataset; demonstrates importance of recent four-class benchmarks.
This work	2026	Masoud + Ishans24 internally; Mendeley 2026 externally	Basic CNN, ResNet50, EfficientNetB0, DenseNet121 + Grad-CAM	Best external weighted F1 = 87.95%	Focuses on leakage control and external validation rather than only internal accuracy.

III. DATASET SOURCES AND JUSTIFICATION

Three public datasets were used. The Masoud Nickparvar Brain Tumor MRI Dataset and Ishans24 Brain Tumor Dataset were used for internal train, validation, and internal test splits. The Mendeley 2026 Brain Tumor MRI Dataset was used only for external validation. The Mendeley source page describes Version 6 as a 2026 dataset containing T1-weighted contrast-enhanced MRI images organized into four classes: glioma, meningioma, pituitary, and no tumor [3]. This class compatibility made it suitable for external validation without changing the problem definition.

The dataset design follows a source-aware rule. Internal data can be used for model fitting, validation-based monitoring, and internal testing. External data must not be used for training, checkpoint selection, or hyperparameter tuning. This rule reduces the risk of optimistic bias and makes the evaluation closer to a realistic domain-shift scenario.

Raw scanning detected 7,200 images from Masoud, 10,560 images from Ishans24, and 11,148 images from Mendeley 2026. After deduplication and source-aware split construction, the final dataset contained 15,721 images: 7,415 training, 1,589 validation, 1,590 internal test, and 5,127 external validation images.

TABLE II: Dataset Sources and Justification

Use	Dataset	Justification
Internal train/validation	Masoud Nickparvar dataset [1]	Large four-class MRI source and a common benchmark for glioma, meningioma, no-tumor, and pituitary classification.
Internal train/validation	Ishans24 dataset [2]	Adds internal diversity and increases class coverage after harmonized class-name mapping.
External validation	Mendeley 2026 Version 6 [3]	Independent source reserved for external generalization testing; compatible four-class class taxonomy.

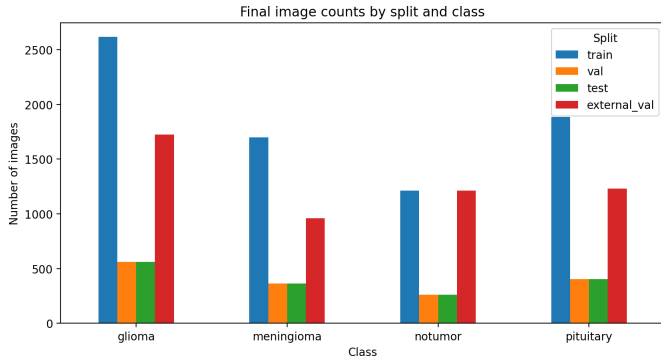


Fig. 1: Final class distribution after preprocessing, deduplication, internal splitting, and external validation construction.

TABLE III: Preprocessing and Leakage-Control Summary

Checkpoint	Count
Raw detected images	28,908
Successfully preprocessed images	28,908
Failed preprocessing	0
Duplicate entries after preprocessing	20,400
Duplicate hash groups after preprocessing	7,213
Internal images after deduplication	10,594
Final external validation images	5,127
Train images	7,415
Validation images	1,589
Internal test images	1,590
Final total images	15,721
Leakage duplicate rows across splits	0
Within-split duplicate rows	0

IV. METHODOLOGY

A. Preprocessing and Leakage Control

The preprocessing workflow converted the raw sources into a unified image classification dataset. The main steps were: dataset scanning, class-name harmonization, image loading, RGB conversion, resizing to 224×224 pixels, hash generation, duplicate detection, duplicate removal, source-aware train/validation/test construction, and final leakage checking. The class labels were standardized as glioma, meningioma, no-tumor, and pituitary.

Duplicate control was treated as a methodological requirement rather than a cleaning detail. The notebook first detected duplicates after preprocessing. Internal duplicates were removed before the internal split. External images were deduplicated within the Mendeley source and then against the internal pool. Final leakage checks found zero duplicate rows across splits and zero within-split duplicate rows.

B. Model Selection, Modification, and Justification

Four models were chosen to create a fair baseline-to-transfer-learning comparison. The Basic CNN was trained from scratch to show how a simple custom model behaves under the same data protocol. ResNet50 was selected because residual skip connections are a strong and widely used

transfer-learning baseline. EfficientNetB0 was selected because compound scaling can produce efficient high-performing models. DenseNet121 was selected because dense connectivity encourages feature reuse and can improve gradient flow.

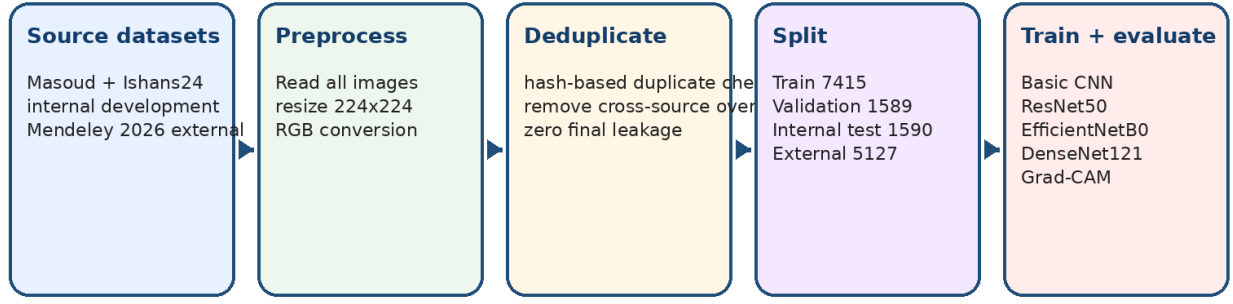
The Basic CNN used four convolutional blocks followed by adaptive average pooling and a two-layer classifier. The transfer-learning models used ImageNet weights where available. Their original classification heads were replaced with a task-specific four-class head: dropout, a linear layer, ReLU activation, another dropout layer, and a final linear layer with four outputs. This modification makes the models compatible with the glioma, meningioma, no-tumor, and pituitary classes.

C. Augmentation, Training, and Fine-Tuning

All images were resized to 224×224 pixels. Training transformations used ImageNet normalization and mild online augmentation: random rotation of 10 degrees, random affine translation of 0.05, scale jitter from 0.95 to 1.05, and contrast jitter of 0.10. Validation, internal test, and external validation used deterministic resize, tensor conversion, and ImageNet normalization only. This separation prevented test-time augmentation from altering the reported evaluation metrics.

Training used PyTorch dataloaders with a batch size of 32. Class weights were computed from the training data and applied to cross-entropy loss to reduce bias from class imbalance. The transfer-learning models used a staged strategy: the feature extractor was initially frozen while the new classifier learned task-specific boundaries; after 10 frozen epochs, the last 40 layers were unfrozen for fine-tuning. Adam optimization used a learning rate of 10^{-3} for Basic CNN and for the frozen stage of transfer models, and 10^{-5} during fine-tuning. Each completed model was trained for up to 100 epochs and the best checkpoint was used for evaluation.

End-to-end preprocessing and evaluation workflow



Key preprocessing outcome: 28,908 raw images were read; 20,400 duplicate entries were identified; final split leakage = 0 duplicate rows.

Fig. 2: End-to-end workflow: public dataset collection, preprocessing, duplicate and leakage control, internal splitting, external validation, model training, evaluation, and Grad-CAM interpretation.

TABLE IV: Model Selection, Modification, and Justification

Model	Training Mode	Modification	Justification
Basic CNN	Scratch training	Four convolutional blocks; adaptive average pooling; dropout-based classifier; four output neurons.	Transparent baseline for measuring whether transfer learning is necessary.
ResNet50	Transfer learning + fine-tuning	Replaced final fully connected layer with dropout-linear-ReLU-dropout-linear four-class head.	Residual connections support deep feature extraction and stable training.
EfficientNetB0	Transfer learning + fine-tuning	Replaced classifier with the same four-class dropout head.	Efficient compound-scaled architecture with strong parameter-performance trade-off.
DenseNet121	Transfer learning + fine-tuning	Replaced classifier with the same four-class dropout head.	Dense feature reuse may support stronger generalization under domain shift.

V. EVALUATION METRICS

The evaluation used accuracy, precision, recall, F1-score, weighted F1, macro F1, and confusion matrices. Accuracy measures total correctness, but it can hide class-specific weakness when classes are imbalanced. Weighted F1 incorporates class support, while macro F1 treats each class equally. For a class c , true positives, false positives, false negatives, and false negatives are denoted as TP_c , FP_c , FN_c , and TN_c . The metrics are:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \quad (1)$$

$$Precision_c = \frac{TP_c}{TP_c + FP_c}, \quad Recall_c = \frac{TP_c}{TP_c + FN_c}, \quad (2)$$

$$F1_c = 2 \times \frac{Precision_c \times Recall_c}{Precision_c + Recall_c}. \quad (3)$$

Confusion matrices were used to identify where models failed across tumor types. Grad-CAM was used as an explainability tool to visualize discriminative image regions contributing to the predicted class [7]. The external validation

metrics were interpreted as the primary indicator of model robustness.

VI. RESULTS AND ANALYSIS

A. Training Curves and Convergence

The separated learning curves in Figs. 3–10 are placed directly inside the results section rather than being grouped at the end of the paper. This makes the convergence evidence easier to read in the two-column IEEE layout. The Basic CNN converged well on internal validation but later showed the largest external gap. ResNet50 reached the most stable high internal validation curve. EfficientNetB0 achieved the highest internal test accuracy, while DenseNet121 showed the best external robustness even though its internal validation curve was not the highest. Therefore, convergence curves are useful for checking training quality, but external validation remains the stronger indicator of deployable robustness.

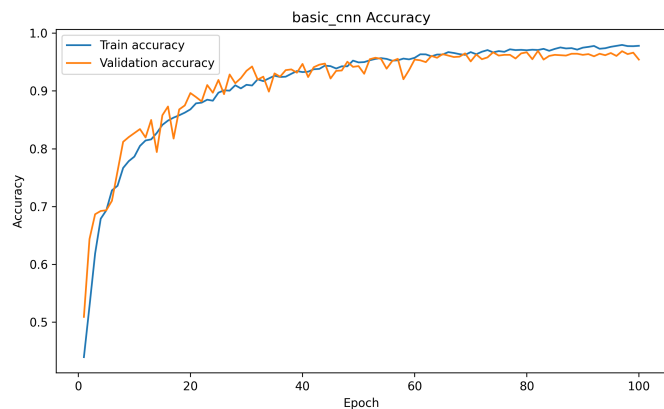


Fig. 3: Basic CNN training and validation accuracy curve.

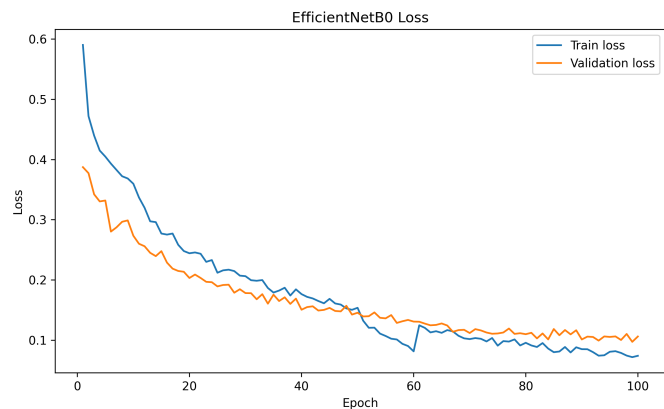


Fig. 6: EfficientNetB0 training and validation loss curve.

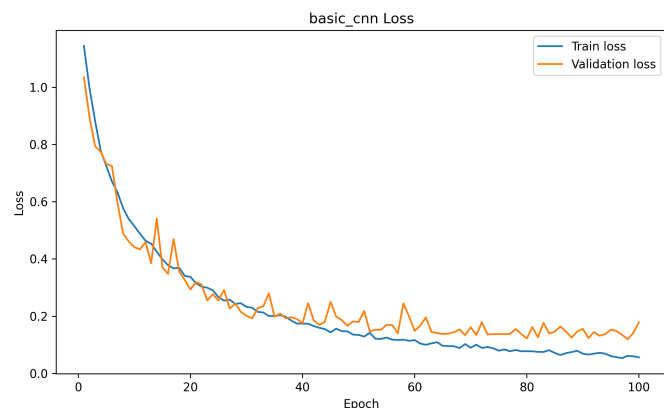


Fig. 4: Basic CNN training and validation loss curve.

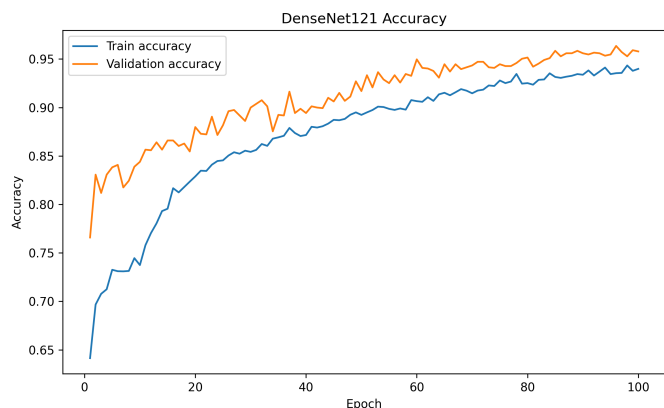


Fig. 7: DenseNet121 training and validation accuracy curve.

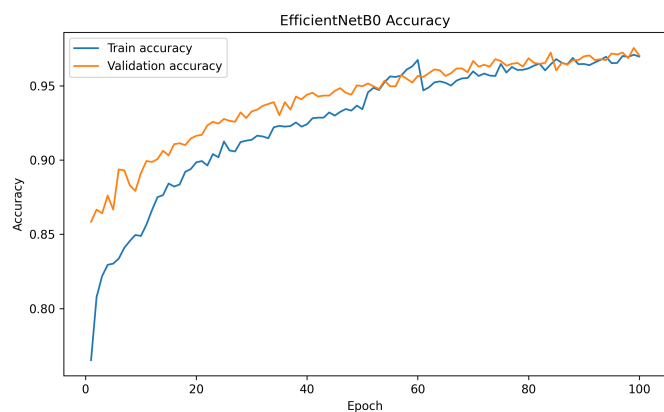


Fig. 5: EfficientNetB0 training and validation accuracy curve.

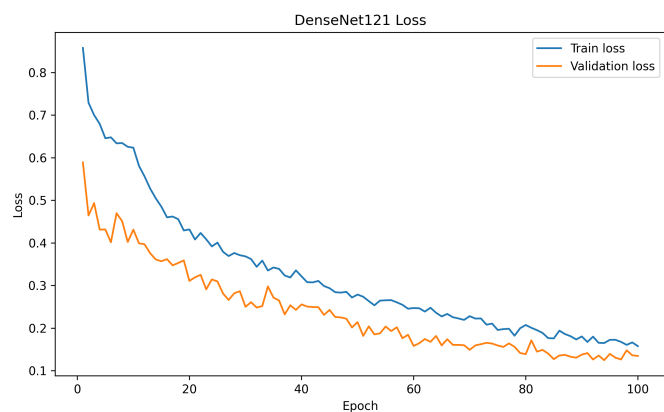


Fig. 8: DenseNet121 training and validation loss curve.

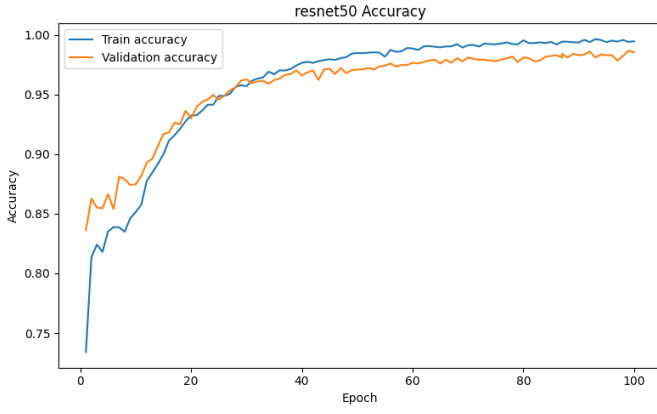


Fig. 9: ResNet50 training and validation accuracy curve.

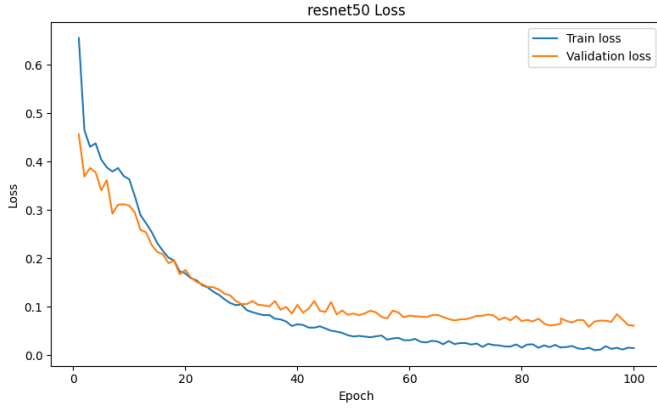


Fig. 10: ResNet50 training and validation loss curve.

B. Overall Model Comparison

Table V compares all four models on internal test and external validation data. EfficientNetB0 achieved the highest internal accuracy of 97.99%, but DenseNet121 achieved the strongest external performance with 87.79% accuracy and 87.95% weighted F1-score. ResNet50 was close behind externally with 87.42% accuracy and 87.58% weighted F1-score. The gap between internal and external performance confirms that the external validation cohort is not redundant: it changes the final model ranking.

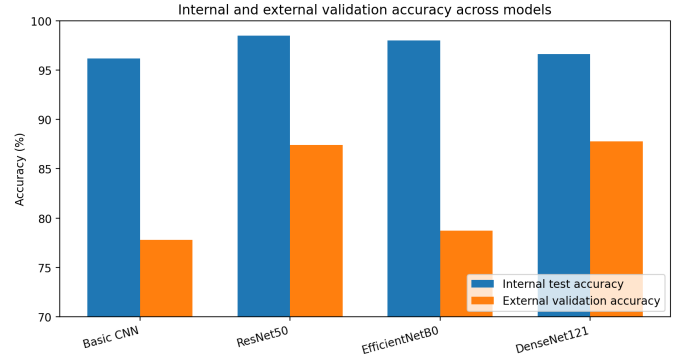


Fig. 11: Internal and external accuracy comparison across the four models.

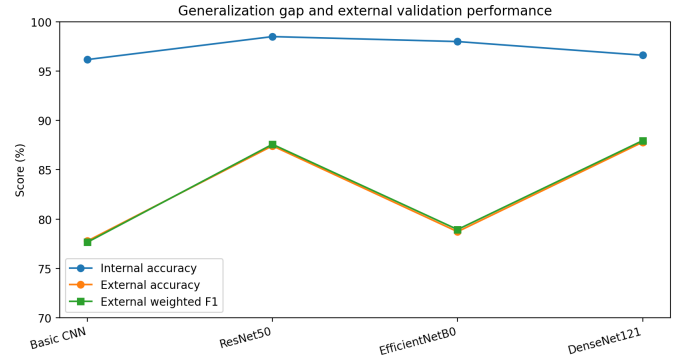


Fig. 12: Generalization gap and external validation performance. DenseNet121 shows the smallest internal-to-external accuracy drop and the highest external weighted F1-score.

TABLE V: Four-Model Internal Test and External Validation Comparison

Model	Int. Acc. (%)	Int. Prec. (%)	Int. Recall (%)	Int. W-F1 (%)	External Acc. (%)	Ext. Prec. (%)	Ext. Recall (%)	Ext. W-F1 (%)
Basic CNN	96.16	96.24	96.16	96.18	77.78	78.52	77.78	77.65
ResNet50	98.49	98.52	98.49	98.49	87.42	89.47	87.42	87.58
EfficientNetB0	97.99	98.01	97.99	97.99	78.72	84.41	78.72	78.94
DenseNet121	96.60	96.77	96.60	96.63	87.79	88.62	87.79	87.95

C. Confusion Matrix Analysis

The confusion matrices are shown as separate figures so that each matrix fits cleanly inside a single IEEE column. On the internal test split, all transfer-learning models show strong diagonal dominance. On the external validation set, errors increase for all models, especially for glioma/no-tumor and pituitary/no-tumor confusion. DenseNet121 correctly classified 1475 glioma, 844 meningioma, 1129 no-tumor, and 1053 pituitary samples in the external cohort. ResNet50 correctly classified 1410 glioma, 877 meningioma, 1200 no-tumor, and 995 pituitary samples. These matrices show that external validation is stricter than internal testing and reveals differences that are hidden by internal accuracy alone.

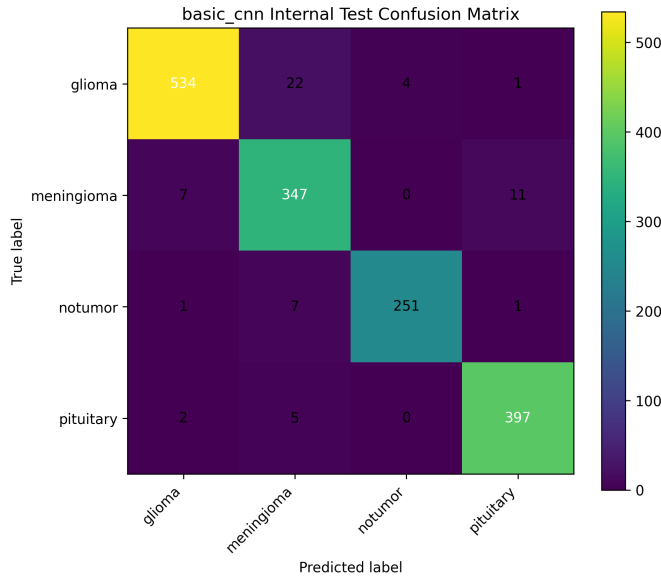


Fig. 13: Basic CNN internal test confusion matrix.

TABLE VI: External Validation Class-Wise F1-Scores

Model	Glioma	Meningioma	No-tumor	Pituitary
Basic CNN	77.90	70.26	84.40	76.42
ResNet50	89.24	86.96	85.14	88.13
EfficientNetB0	82.49	75.38	78.27	77.41
DenseNet121	89.10	80.34	90.00	90.27

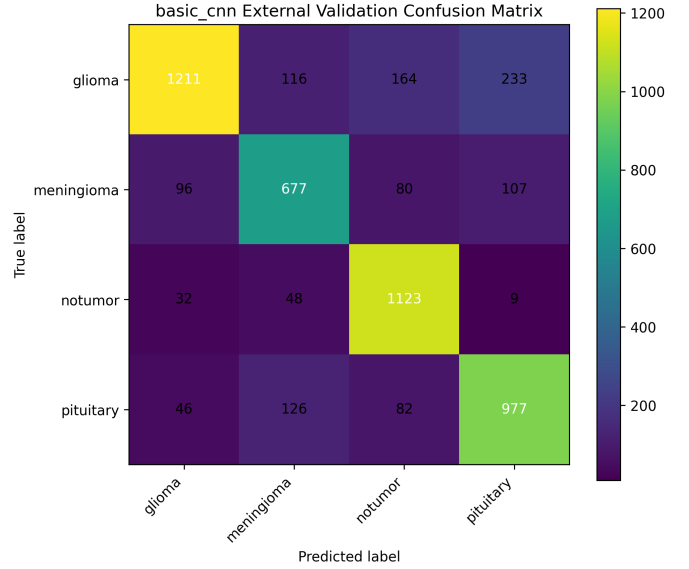


Fig. 14: Basic CNN external validation confusion matrix.

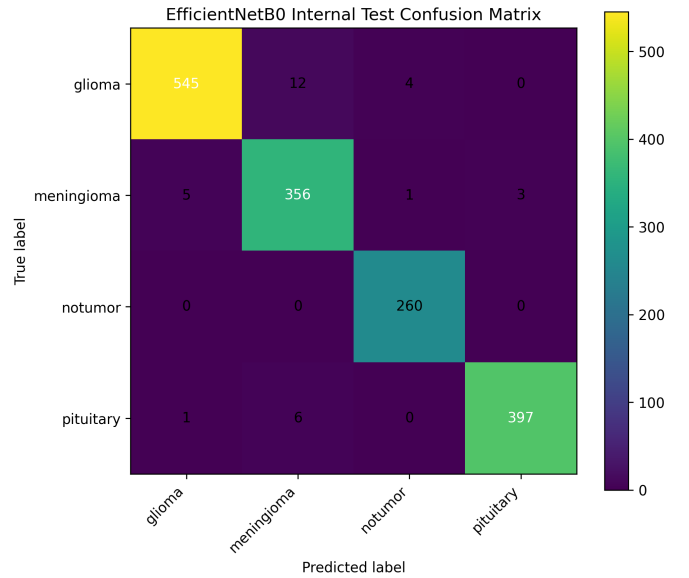


Fig. 15: EfficientNetB0 internal test confusion matrix.

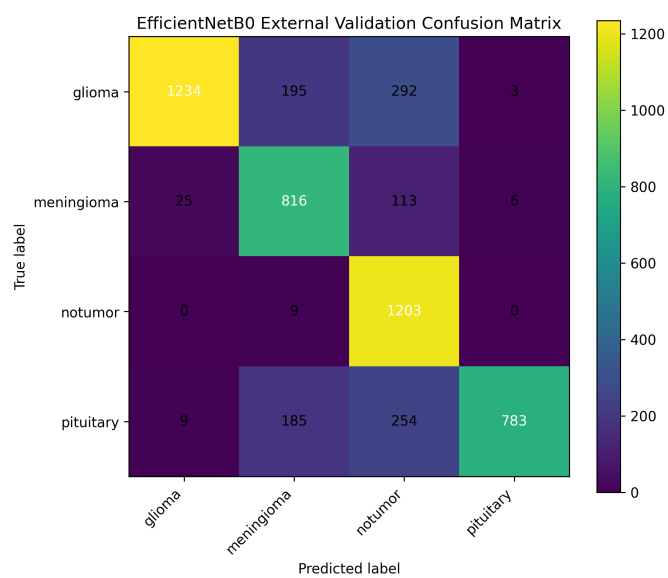


Fig. 16: EfficientNetB0 external validation confusion matrix.

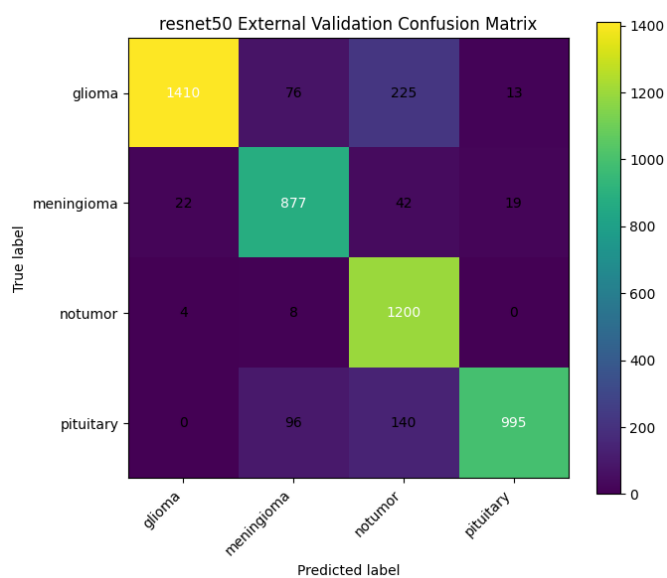


Fig. 18: ResNet50 external validation confusion matrix.

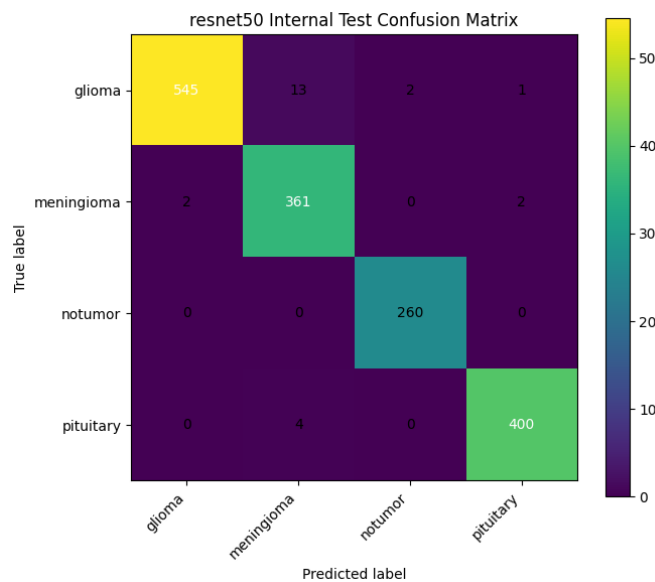


Fig. 17: ResNet50 internal test confusion matrix.

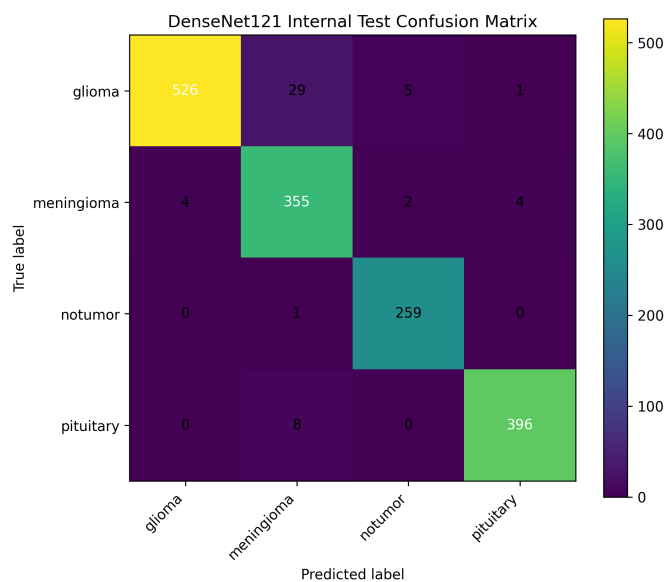


Fig. 19: DenseNet121 internal test confusion matrix.

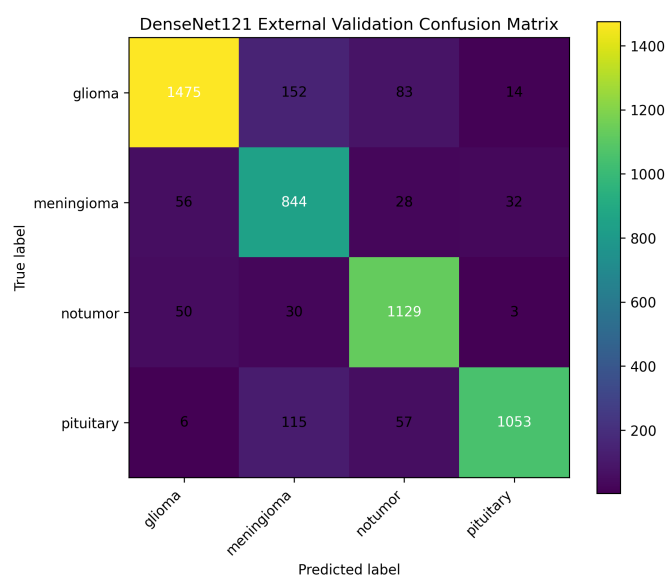


Fig. 20: DenseNet121 external validation confusion matrix.

D. External Validation Interpretation

The external validation set changed the model ranking. EfficientNetB0, despite leading internally, dropped to 78.72% external accuracy. DenseNet121 produced the best external robustness and had the smallest accuracy gap. This suggests that DenseNet-style feature reuse transferred better across the source shift between the internal Kaggle cohort and the Mendeley 2026 validation cohort. ResNet50 also generalized well and achieved the highest external weighted precision among the models. The Basic CNN remained a useful baseline but had limited external robustness compared with the transfer-learning models.

E. Grad-CAM Visualization and Interpretation

Grad-CAM visualizations were generated after evaluation to inspect whether model predictions were supported by meaningful image regions. Instead of a combined CAM montage, representative examples are placed as separate figures. Correct glioma, meningioma, and pituitary cases usually concentrate heatmap intensity around tumor-proximal areas. No-tumor examples often show broader attention because the absence of tumor is not localized to one lesion. Wrong external predictions frequently show high activation around ambiguous or non-target regions, which supports the confusion-matrix finding that source shift affects both localization and classification.

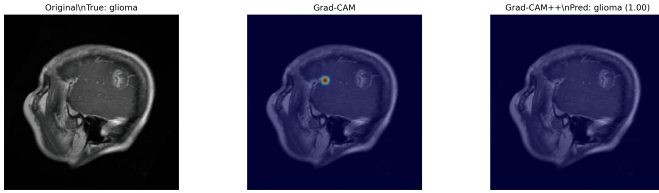


Fig. 21: Basic CNN external Grad-CAM: correct glioma prediction.

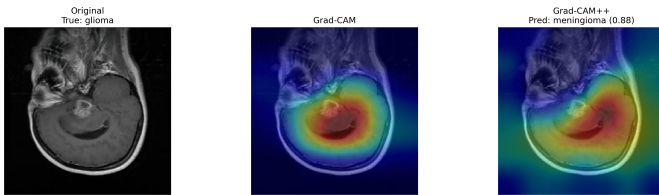


Fig. 22: EfficientNetB0 external Grad-CAM: true glioma predicted as meningioma. The wrong example helps interpret external-domain errors.

Grad-CAM should be interpreted as a model-auditing aid rather than a clinical explanation by itself. A strong heatmap over a lesion-like region increases confidence that the model is using plausible image evidence, but it does not replace radiologist verification. The value of Grad-CAM in this study is that it complements numerical metrics by revealing whether high-confidence errors are visually plausible or attention-dispersed.

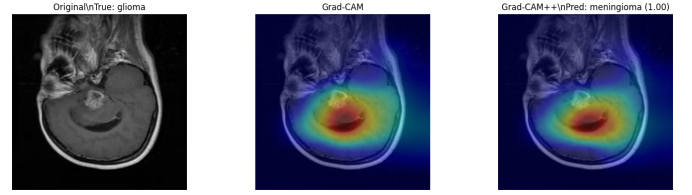


Fig. 23: ResNet50 external Grad-CAM: high-confidence wrong glioma-to-meningioma example.

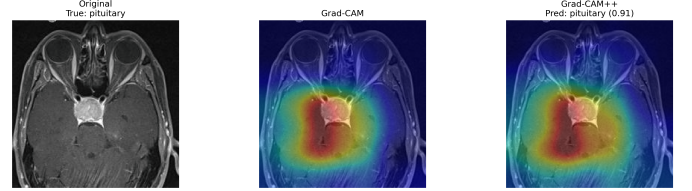


Fig. 24: DenseNet121 external Grad-CAM: correct pituitary prediction with localized activation near the sellar region.

VII. DISCUSSION

The results lead to four main findings. First, duplicate and leakage control are essential. The preprocessing pipeline found 20,400 duplicate entries after preprocessing, which means that naive splitting could have inflated performance. Second, external validation is necessary because internal test metrics alone would select EfficientNetB0 as the best model, whereas external validation identifies DenseNet121 as the most robust. Third, transfer learning is beneficial: the scratch CNN achieved good internal performance but weaker external performance. Fourth, explainability adds qualitative value by revealing where models focus during prediction; the separated Grad-CAM figures make these examples readable within the IEEE two-column format.

The strongest final model under external validation was DenseNet121. Its external weighted F1-score of 87.95% and smaller generalization gap make it the recommended model for this evidence set. ResNet50 is a close alternative because it achieved 87.58% external weighted F1-score and high external precision. EfficientNetB0 is internally strong but less stable externally, indicating that model ranking depends on the evaluation distribution.

VIII. LIMITATIONS AND FUTURE WORK

This study uses public image-level datasets. Patient-level metadata, scanner protocols, slice positions, tumor grade, segmentation masks, and radiologist annotations were not available in the extracted project artifacts. Therefore, the results should be interpreted as public-dataset classification evidence rather than a clinically deployable diagnostic system. Future work should use patient-level splits, multi-institutional clinical cohorts, calibration analysis, uncertainty estimation, segmentation-assisted classification, and radiologist reader studies. Additional explainability methods such as Grad-CAM++ or Score-CAM can be added as bonus explainability

variants, and a multi-task model could jointly classify tumor type and estimate lesion localization if bounding-box or segmentation annotations become available.

IX. CONCLUSION

This paper presented a structured and externally validated brain tumor MRI classification study using three public datasets and four deep learning models. The preprocessing pipeline converted 28,908 detected images into 15,721 leakage-controlled final images. EfficientNetB0 achieved the strongest internal test accuracy, but DenseNet121 achieved the best external validation performance with 87.79% accuracy and 87.95% weighted F1-score. The change in model ranking under external validation confirms the central message of this work: credible medical imaging benchmarks must report duplicate-control, internal metrics, external validation, confusion matrices, and interpretable visualizations together.

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